

Reports on Self-Driving Cars Analysis

Executive Summary & Investment Decision

Based on the analysis of booking demand, maintenance costs, and accident data from the MonCity data warehouse, the self-driving car project shows potential for continued investment, but the budget should be allocated carefully. The data records 10,000 bookings and approximately \$303,000 in maintenance costs. However, several maintenance, safety, and demand-related issues should be addressed to improve the project's efficiency and value.

Key Findings & Data Warehouse Evidence

1. Vehicle Type: Focus on People Movers

- **Maintenance efficiency:** Buses have the highest maintenance cost per booking at approximately \$34 and around 8.5 bookings between maintenance events. People Movers have a lower maintenance cost per booking of approximately \$29 and around 8.8 bookings between maintenance events.
- **Accident severity:** People Movers have a higher proportion of No Damage and Minor Damage accidents, while Mini Buses have a higher proportion of Moderate and Severe Damage accidents.
- **Recommendation:** Consider prioritising People Movers in future fleet expansion while reviewing the safety and maintenance performance of Mini Buses before increasing their share of the fleet.

2. Main Technical Problem: Error002 — Low Battery

- **Main error:** Error002, representing a Low Battery issue, is the most common accident-related error, with 373 accidents, accounting for approximately 37% of the 1,000 recorded accidents.
- **Main locations:** Error002 occurs most frequently in Zone B (104 accidents), followed by Zone A and Zone C (90 accidents each).
- **Recommendation:** Prioritise engineering improvements for the Low Battery issue, particularly in Zones A, B, and C. Possible measures include earlier battery warnings, real-time battery monitoring, and preventive charging or battery replacement before critical battery levels are reached.

3. Maintenance Cost Distribution

- **Cost concentration:** A significant share of allocated maintenance costs is associated with Boeing Phantom Works (CE02) and Palo Alto Research Center (CE04).
- **Recommendation:** Review maintenance activities, resource allocation, and costs at these research centres to identify opportunities to reduce unnecessary expenditure while maintaining vehicle reliability.

4. User Groups & Demand Patterns

- **Main users:** Middle-aged adults are the largest user group, accounting for approximately 51% of bookings, followed by young adults at 41%.
- **Top faculties:** Information Technology (approximately 3,000 bookings) and Engineering (approximately 2,400 bookings) have the highest fleet usage.

- Monthly demand: Fleet usage is generally consistent throughout the year, with relatively lower demand observed in February and April.

Recommendations for Management

1. **Fleet Allocation:** Consider allocating a greater proportion of future fleet investment to People Movers, based on their lower maintenance cost per booking and more favourable accident-severity profile. Before expanding Mini Bus purchases, management should review their higher proportion of severe-damage accidents.
2. **Improve Battery Management:** Since Error002 (Low Battery) accounts for approximately 37% of recorded accidents, introduce stricter battery management measures. These could include earlier battery-warning thresholds, battery-level monitoring during trips, and scheduled charging or replacement before vehicles reach critical levels. Zones A, B, and C should receive particular attention.
3. **Review Maintenance Costs:** Conduct a detailed review of maintenance activities and resource allocation at CE02 and CE04, with particular attention to identifying high-cost maintenance activities and opportunities to improve resource efficiency.
4. **Increase Bookings During Lower-Demand Periods:** Introduce targeted campaigns during February and April, particularly among lower-usage faculties such as Business and Economics. Possible strategies include faculty-specific awareness campaigns, repeat-booking incentives, and targeted communications to increase fleet utilisation during these periods.